1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In NMIBC, does the prognostic impact of ADNP-high CTCs on progression-free interval depend on the presence of TLS-like lymphoid aggregates and the patient's HDL-C level? Mechanistically, the absence of TLSs and low HDL-C facilitate ADNP-driven tumor clones to evade local antigen presentation, leading to faster progression; clinically, ADNP-high CTCs would predict shorter PFI specifically in TLS-low and low-HDL-C patients, defining a high-risk subgroup for intensified surveillance or lipid-immune–modulating interventions.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	ADNP, HDL-C
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

ADNP: ADNP mRNA detected in CTCs (liquid biopsy transcript axis) because it provides a tumor-in-blood readout of aggressive clones; (b) HDL-C as a systemic metabolic biomarker because lipid status can shape immune function and tumor dissemination; (c) the AI/MSI/accessibility path was considered mechanistically (as a marker of genomic instability that might interact with immune organization) but deprioritized from the core hypothesis to maintain focus.

HDL-C: HDL-C as a systemic metabolic biomarker because lipid status can shape immune function and tumor dissemination; (c) the AI/MSI/accessibility path was considered mechanistically (as a marker of genomic instability that might interact with immune organization) but deprioritized from the core hypothesis to maintain focus.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CTCs	Supporting modifier from knowledge graph
Secondary	MSH6	Supporting modifier from knowledge graph

Secondary	MSI	Supporting modifier from knowledge graph
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	TMB	Supporting modifier from knowledge graph
Secondary	pZVR	Supporting modifier from knowledge graph
Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CTCs: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MSH6: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.